**Project Design Phase-I**
**Solution Architecture**

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| Date | 26 October 2026 |
| Team ID | Gyanvi Jain |
| Project Name | Formula 1 Race Intelligence Dashboard |
| Maximum Marks | 4 Marks |

### Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

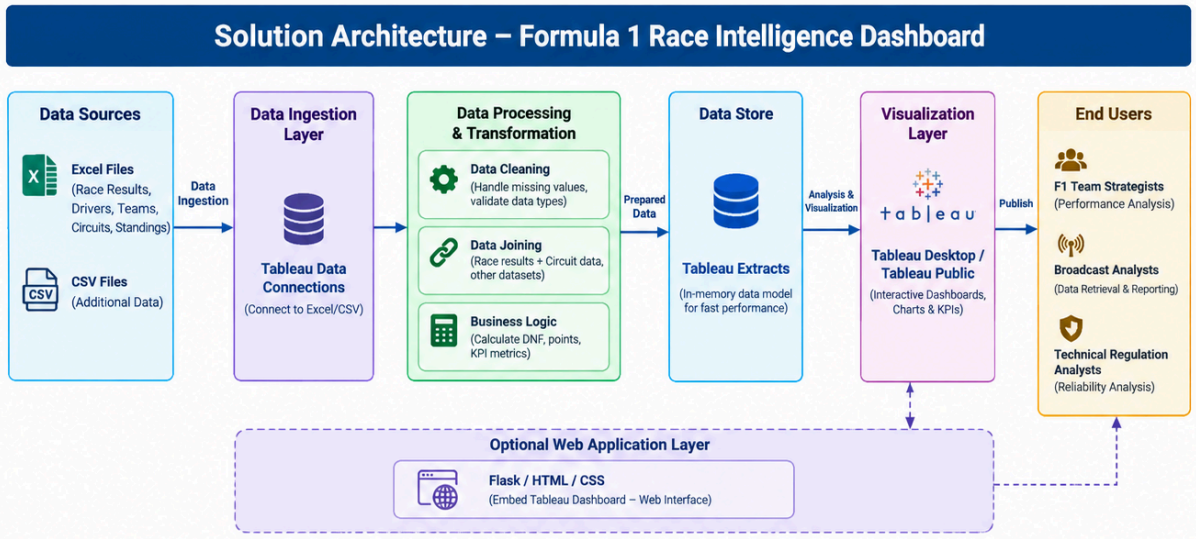
**Project Name: Formula 1 Race Intelligence Dashboard**

**The solution architecture describes how Formula 1 data moves from source files through data preparation and analysis into an interactive dashboard. The system uses Tableau for data processing and visualisation, Tableau Public for publishing, and an optional Flask web application for embedding the dashboard.**

## Architecture Workflow

1. Data collection: Formula 1 datasets are provided through Excel and CSV files.
2. Data ingestion: Tableau connects to the source files.
3. Data processing: Data is cleaned, validated, joined, and transformed.
4. Data storage: Prepared data is made available through the Tableau data source or extracts.
5. Analysis and visualisation: Tableau generates KPIs, charts, filters, and interactive dashboards.
6. Publishing: The dashboard is published using Tableau Public.
7. User access: F1 analysts and other users explore the dashboard directly or through an optional Flask web application.

## Solution Architecture Diagram:




**Component Description Table**

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Data Sources | Provide raw Formula 1 data including race results, drivers, teams, circuits and championship standings. | Excel, CSV |
| Data Ingestion Layer | Connects to source files and loads data into the analytical environment. | Tableau Data Connections |
| Data Processing & Transformation | Cleans data, handles missing values, joins datasets and applies business logic to prepare data for analysis. | Tableau Prep, Tableau Desktop, Calculated Fields, SQL (optional) |
| Data Store | Stores the prepared data in a format optimized for analysis and performance. | Tableau Extracts |
| Visualization Layer | Creates interactive charts, dashboards, KPIs and visualisations for insights and reporting. | Tableau Desktop / Tableau Public |
| End Users | Access and analyse the data to make informed decisions and prepare reports. | Web Browser (Tableau Public), Flask (optional) |
| Web Application Layer (Optional) | Embeds the Tableau dashboard in a web interface for broader access and integration. | Flask, HTML, CSS |

## Component Description Table

| Component Name | Description / Role in Architecture | Technology Used |
|---|---|---|
| Data Sources | Provides Formula 1 race results, driver details, team information, circuit data, and championship standings. | Excel, CSV |

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|---|---|---|
| Data Ingestion Layer | Connects the source datasets to Tableau for analysis. | Tableau Data Connections |
| Data Processing & Transformation | Handles missing values, validates data types, joins race results with circuit data, and calculates DNF counts and cumulative points. | Tableau Desktop, Calculated Fields |
| Data Store | Holds the prepared data in a format suitable for analysis and visualisation. | Tableau Data Source / Extracts |
| Business Analysis Layer | Calculates KPIs and analyses driver performance, team standings, tyre usage, reliability, and podium achievements. | Tableau Calculated Fields, Tableau Worksheets |
| Visualisation Layer | Displays interactive charts, KPI cards, filters, parameters, and the final dashboard. | Tableau Desktop, Tableau Public |
| Web Application Layer (Optional) | Embeds the published Tableau dashboard in a web interface. | Flask, HTML, CSS |
| End Users | Access and analyse dashboard insights to compare driver and team performance and explore race trends. | Web Browser, Tableau Public |